

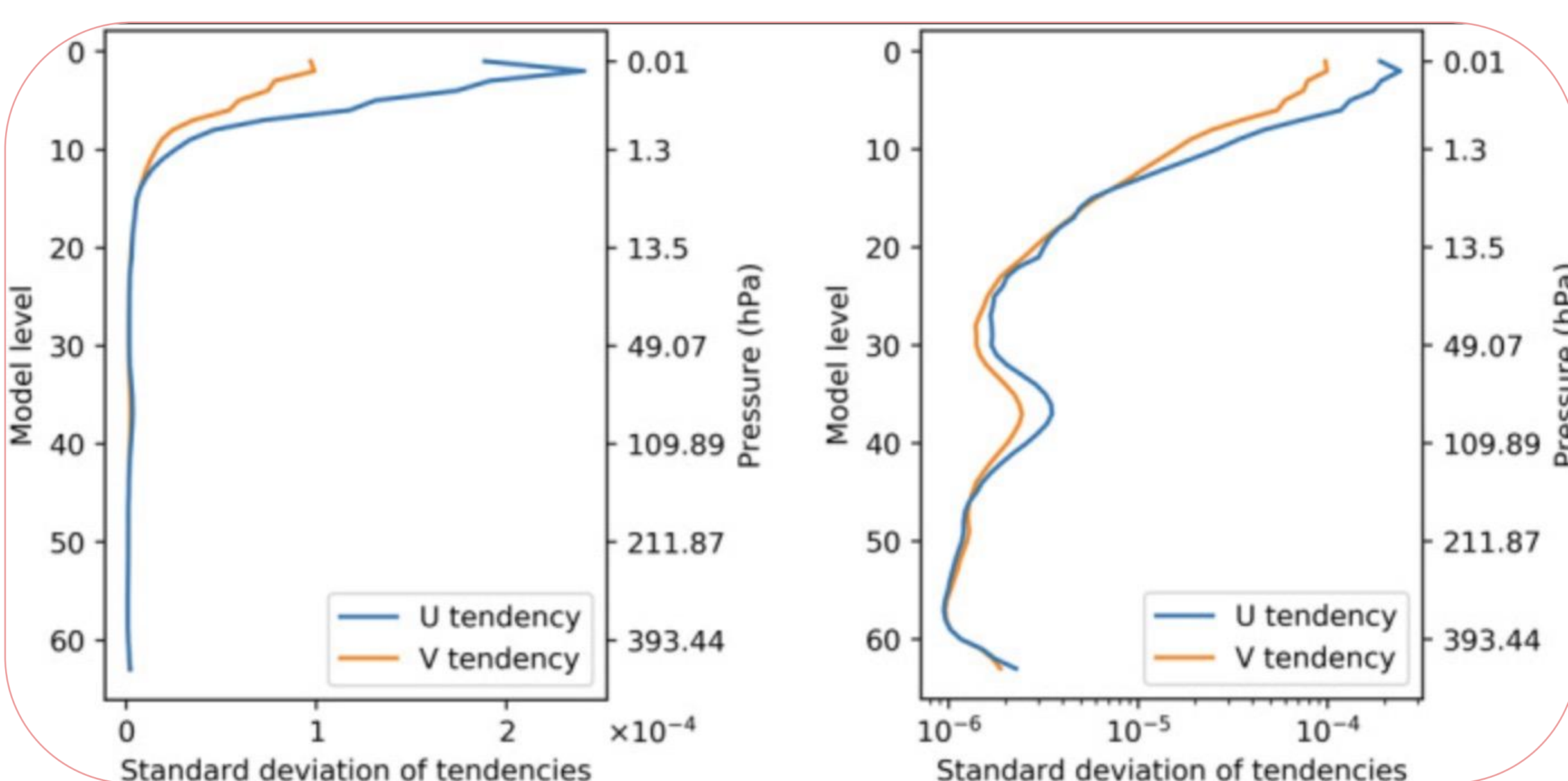


PREDICTING NATURAL DISASTERS EARLIER: USING ATMOSPHERIC WAVES TO FORECASTING EARTHQUAKES, FLOODS, AND STORMS IN INDIA

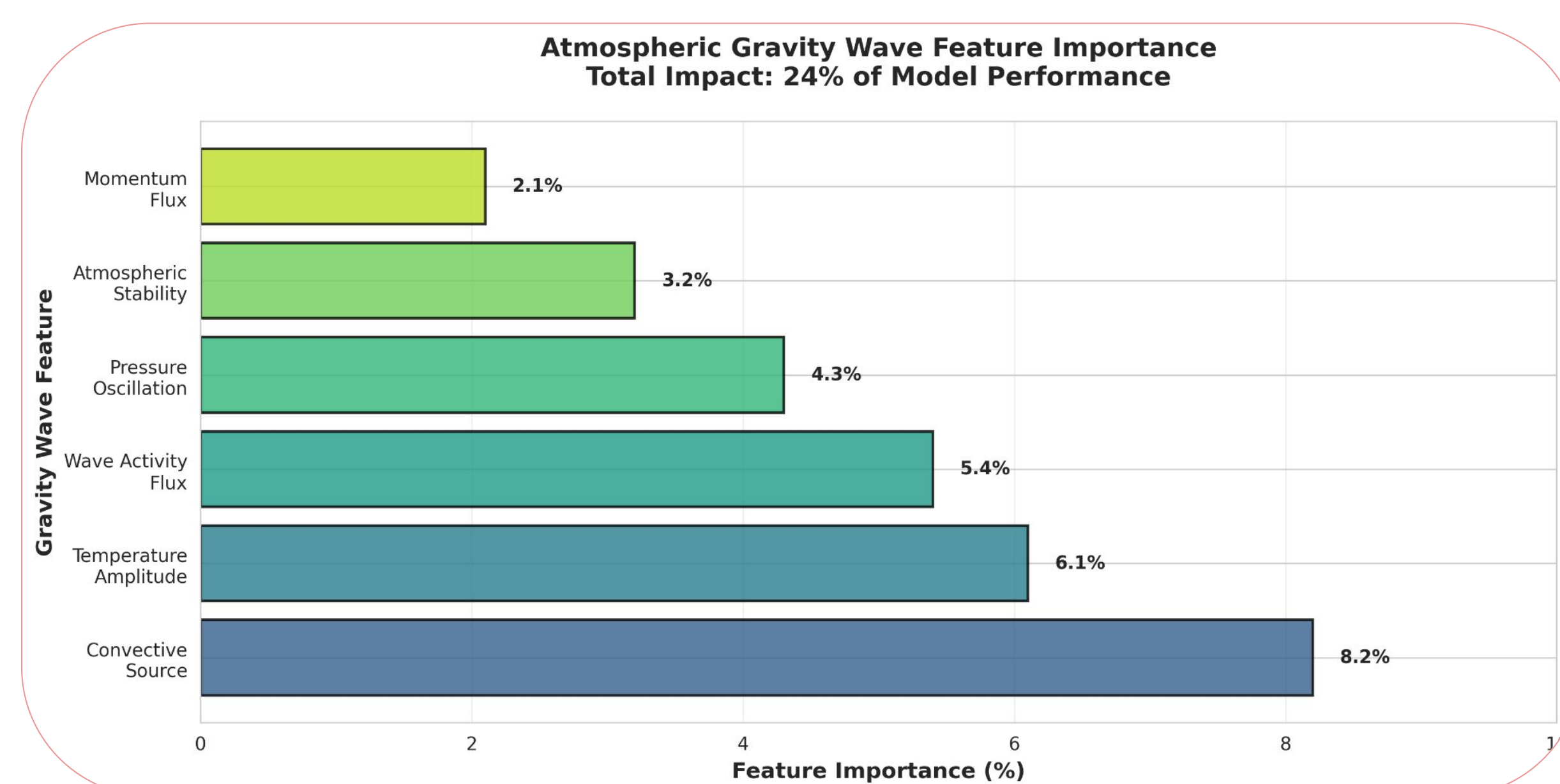
SPIRAL: Structured Physics-Informed Representation-Augmented Learning from Atmospheric Gravity Wave Signals for Multi-Hazard Prediction in India and Real-Time Web and Mobile Risk Alerts

Atharv Malpani, Class of 2031 & Johnathan Ahdout, Class of 2029

February 20, 2026



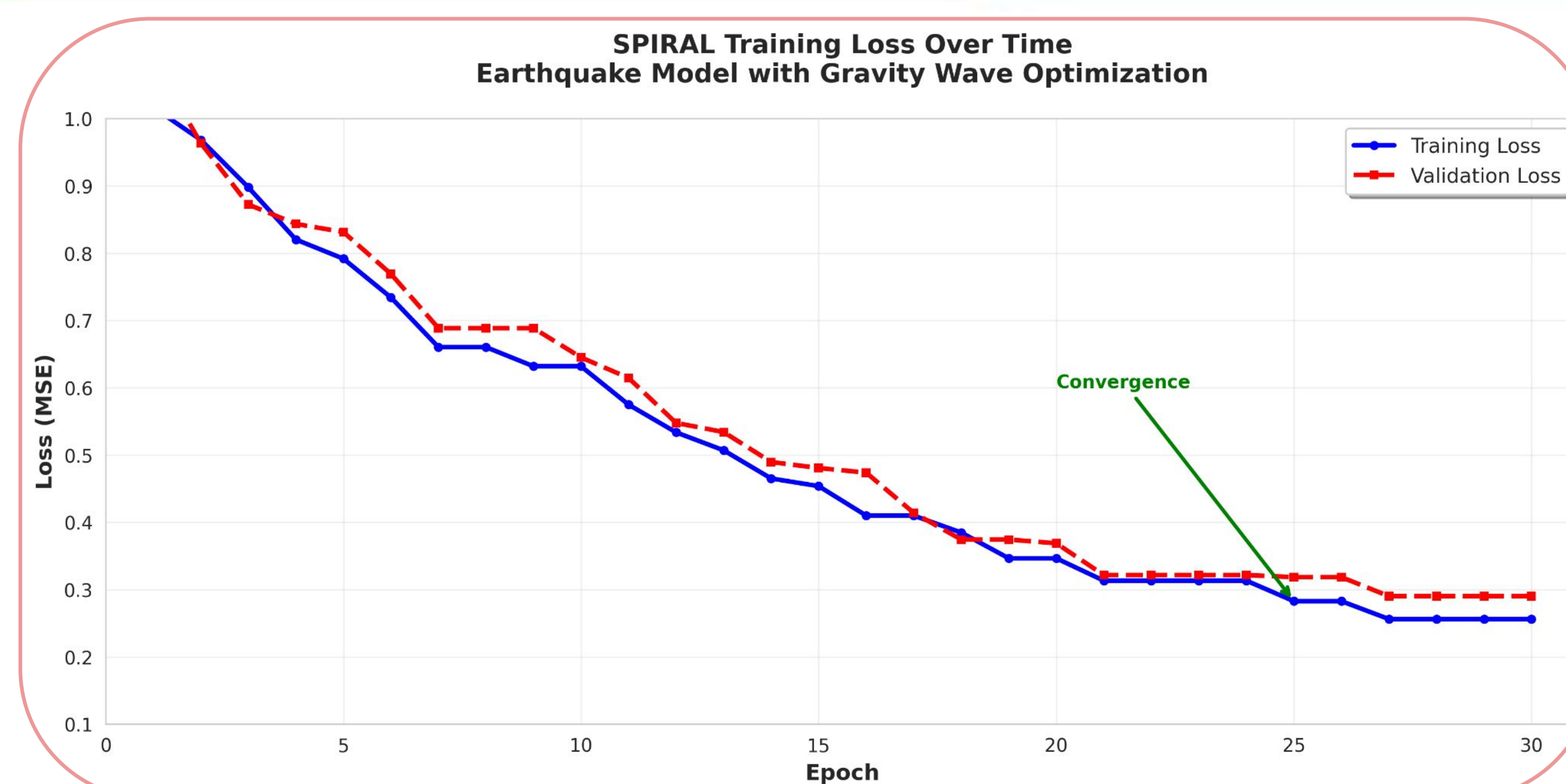
Chantry, M., Hatfield, S., Dueben, P., Polichtchouk, I., & Palmer, T. (2021). Machine Learning Emulation of Gravity Wave Drag in Numerical Weather Forecasting. *Journal of advances in modeling earth systems*, 13(7), e2021MS002477. <https://doi.org/10.1029/2021MS002477>



- Data Sources**
- USGS Earthquake Catalog
 - 702 real earthquake events
 - Meteorological observations
 - Data: 2023-2024 (India focus)

Key Innovation

Atmospheric Gravity Waves are ripples in the atmosphere that appear **hours before** severe weather events. By detecting these pre-storm signals, SPIRAL can predict disasters earlier than traditional methods.



<https://github.com/JohnathanAhdout/SPIRAL>

